

Tianyi Ouyang

Ph.D.in Physics

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RESEARCH INTERESTS

Optical spectroscopy of two-dimensional semi-conductive materials (mainly Transition Metal Dichalcogenides)

EDUCATION

University of California, Riverside, Riverside, California, USA

September 2019 to June 2025

Ph.D. in Physics

Advisors: Chun Hung Lui (Chair); Nathaniel M. Gabor (Committee); Yong-Tao Cui (Committee)

University of Science and Technology of China, Hefei, Anhui, China

August 2015 to June 2019

B.S. in Physics (Theoretical physics)

Advisor: Zhenyu Zhang

AWARD

Dissertation Completion Fellowship Award

Jan 2025 to June 2025

Outstanding Teaching Assistant Award

Sept 2024 to June 2025

USTC Outstanding Student Scholarship Silver Award

2016

CONFERENCE

Presenter in American Physical Society March Meeting 2025

March 2025

Enhancing resonant second harmonic generation in bilayer WSe₂ by layer-dependent exciton-polaron effect

Presenter in American Physical Society March Meeting 2024

March 2024

Electric-field-switchable exciton polarons involving Γ -valley and K-valley Fermi sea in trilayer WSe₂

WORKING EXPERIENCE

Postdoctoral Research Associate in Carnegie Mellon University

Sept 2025 to Present

PI: Su-Fei Shi

Teaching Assistant in University of California, Riverside

Sept 2019 to Dec 2024

SELECTED COURSES IN PH.D.

- Classical Mechanics
- Electromagnetic Theory
- Quantum Mechanics
- Methods of Theoretical Physics
- Thermodynamics and Statistical Mechanics
- Condensed Matter Physics
- Nonlinear Optics

RESEARCH SKILLS

- **Programming:** Python, Labview
- **Software:** Igor Pro, Origin, Mathematica, MatLab

- **Sample fabrication:** Dry transfer, E-Beam Lithography, E-Beam Evaporation, Atomic Force Microscopy, Wire Bonder, Glove Box
- **Optics:** PL and reflectance optical setup and measurement, Montana/Janis/AttoDry cryostat experience

SELECTED PUBLICATIONS

Nature Physics accepted (2026)

Soonyoung Cha[†], Zhaoran Xu[†], **Tianyi Ouyang[†]**, Hongyu Yao, Raj Kumar Paudel, Takashi Taniguchi, Kenji Watanabe, Andrew Y. Joe, Yia-Chung Chang, Nathaniel M. Gabor and Chun Hung Lui

Hybridization-revealed interlayer 2p exciton state in bilayer MoS₂

Phys. Rev. Lett. 136 (096903) (2026) (Editor's Suggestion)

Tianyi Ouyang[†], Erfu Liu[†], Soonyoung Cha[†], Raj Kumar Paudel, Yiyang Sun, Zhaoran Xu, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor, Yia-Chung Chang and Chun Hung Lui

Brightening Interlayer Excitons by Electric-Field-Driven Hole Transfer in Bilayer WSe₂

arXiv: 2601.11914 (2025) (Nature Physics under reviewing)

Erfu Liu, Matthew Wilson, Jenny Hu, Alexandra Zimmerman, Amal Mathew, **Tianyi Ouyang**, Ao Shi, Takashi Taniguchi, Kenji Watanabe, Tony. F. Heinz, Yia-Chung Chang, Chun Hung Lui

Exciton-polaron Umklapp Scattering in Wigner Crystals

Science 390, 241-241(2025)

Tianyi Ouyang[†], Su-Fei Shi

Quantum Trions in Equilibrium

Nano Letters 25, 4, 1459-1465 (2025)

Tianyi Ouyang[†], Soonyoung Cha[†], Yiyang Sun, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor and Chun Hung Lui

Electrically Switching Ferroelectric Order in 3R MoS₂ Layers

Nano Letters 24, 46, 14847-14853 (2024)

Soonyoung Cha[†], **Tianyi Ouyang[†]**, Takashi Taniguchi, Kenji Watanabe, Nathaniel M. Gabor and Chun Hung Lui

Enhancing Resonant Second Harmonic Generation in Bilayer WSe₂ by Layer-dependent Exciton-polaron Effect

CONTACT REFERENCE

Chun Hung Lui(PhD Advisor; Professor) University of California, Riverside

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Nathaniel Gabor(PhD Committee; Collaborator; Professor) University of California, Riverside

nathaniel.m.gabor@gmail.com

Yia-Chung Chang(Collaborator; Professor) Research Center for Applied Sciences, Academia Sinica, Taipei

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Soonyoung Cha(Collaborator; Professor) Sookmyung Women's University

scha@sookmyung.ac.kr

Erfu Liu(Collaborator; Professor) Nanjin University

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